



Quiz -1
Name:

CSE341: Microprocessor
Section:

Marks: 15

Duration: 20 minutes
ID:

Question 1: A smart thermostat is being developed for home use. It needs to monitor room temperature and activate a heater or cooler accordingly. The system must operate on battery power, have minimal hardware cost, and only perform basic control logic in a loop. As a consultant, choose whether to use a μP or a μC and give reason. **[1 Mark]**

- Microcontroller (μC)
- Write reason.

Question 2: A handheld gaming console is being designed to run modern games with detailed graphics, support controllers and touch input, and allow frequent game updates. The device should maintain good performance while optimizing battery life for extended play sessions. As a consultant, choose whether to use a μP or a μC and give reason. **[1 Mark]**

- Microprocessor (μP)
- Write reason.

Question 3: What are the main components of the CPU and their functions? **[3 Mark]**

- Control Unit and Instruction Decoder
- ALU
- Registers

Write their function in one line.

Question 4 : Write two differences between RAM and ROM. **[2 Mark]**

- RAM allows both read and write operations. But ROM allows only read operations.
- RAM is volatile; it loses its data when the power is turned off. But ROM is non-volatile; it retains data even when the power is turned off.

Question 5: Short-Answer Questions [2 Marks]

1. What is the role of the Program Counter in a computer system?

- Holds the memory address of the next instruction to be executed

2. Which part of the CPU would most likely handle calculation?

- Arithmetic Logic Unit (ALU)

Question 6: Deduce the size of the address bus if the total size of the memory is 4 GB. **[3 Mark]**

- Total memory size is 4 GB which is $= 4 \times 1024 \times 1024 \times 1024$ bytes $= 4294967296$ bytes
- Address Bus Size $= \log_2(\text{Memory Size})$
- Address Bus Size $= \log_2(4294967296) = 32$ bits
- So, the address bus size is 32 bits.

Question 7: Deduce the 5th largest and the 5th smallest possible **segment numbers** and **logical addresses** (segment: offset pair) for the given physical address: **63583h**. **[3 Mark]**

- Given Physical Address $= 63583\text{h}$
- $63583\text{h}/10$ [Not divisible by 10 in decimal] $\approx 63580\text{h}$ (Nearest lower number divisible by 10 in decimal)
- 1st largest $= 63580\text{h}$
- 2nd largest $= 63570\text{h}$
- 3rd largest $= 63560\text{h}$
- 4th largest $= 63550\text{h}$
- 5th largest $= 63540\text{h}$
- So, 5th largest Segment number $= 63540\text{h}/10 = 6354\text{h}$ and Offset $= (63583\text{h} - 63540\text{h}) = 0043\text{h}$
- **5th largest logical address = 6354 : 0043**
- Again, Given Physical Address $= 63583\text{h}$
- Smallest valid address $= (63583\text{h} - \text{FFFFh}) = 53584$
- $53584\text{h}/10$ [Not divisible by 10 in decimal] ≈ 53590 (Nearest higher number divisible by 10 in decimal)
- 1st smallest $= 53590\text{h}$
- 2nd smallest $= 535\text{A}0\text{h}$
- 3rd smallest $= 535\text{B}0\text{h}$
- 4th smallest $= 535\text{C}0\text{h}$
- 5th smallest $= 535\text{D}0\text{h}$
- So, 5th smallest Segment number $= 535\text{D}0\text{h}/10 = 535\text{Dh}$ and Offset $= (63583\text{h} - 535\text{D}0\text{h}) = \text{FFB}3\text{h}$
- **5th smallest logical address = 535D : FFB3**